Engsci 721 Inverse Problems and Learning From Data

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Lecture 11: Iterative regularisation

Problem 1

Suppose that a matrix A is too large to store or factorise, but that you can efficiently evaluate the matrix-vector products Av and $A^T w$ for vectors v and w of compatible dimensions.

- Explain what is meant by a *matrix-free* iterative method in this setting.
- Explain why such a method might be preferable to computing a pseudoinverse or SVD of A.
- Give one further reason why iterative methods are useful for nonlinear inverse problems.

Problem 2

Consider the least squares normal equations

$$A^T Ax = A^T y$$

and the least squares objective

$$J(x) = \frac{1}{2}\|Ax - y\|^2.$$

- Reformulate the normal equations as a fixed point problem and hence derive the Landweber iteration

$$x_{k+1} = x_k + \alpha A^T (y - Ax_k),$$

where $\alpha > 0$ is a step size.

- Using matrix calculus, show that the same scheme is gradient descent applied to $J(x)$.
- Which matrix-vector products are needed at each iteration? If the sequence converges, what equation must its limit satisfy?

Problem 3

Consider the following simple diagonal forward model:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}, \qquad x^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad Ax^* = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}.$$

Suppose that the measured data are

$$y = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = Ax^* + \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

Starting from $x_0 = 0$, apply Landweber iteration with $\alpha = 1$.

- Compute x_1, x_2, and x_3. Also compute the unregularised solution $A^{-1}y$, to which the iteration eventually converges. Compare $A^{-1}y$ with the true value x^* and explain why they are not the same, even though $A(A^{-1}y) = y$.
- Compute the data misfits $\|Ax_k - y\|$ and the parameter errors $\|x_k - x^*\|$ for $k = 1, 2, 3$. Which of these iterates is closest to x^*? Does the iterate with the smallest data misfit give the best estimate of x^*? Hint: after the first update, only the second component contributes to either norm.
- Compare how quickly the first and second components approach their respective values in the unregularised solution. Relate the difference to the singular values of A. Hint: since A is diagonal with positive entries, $U = V = I$ gives an SVD and the diagonal entries are its singular values.
- Use your results to explain how this example illustrates *semi-convergence*: the data misfit continues to improve, while the error relative to x^* first decreases and then increases. Explain why the stopping iteration k can be regarded as a regularisation parameter and describe one possible stopping rule.

Lecture 12: Stochastic/minibatch gradient descent

Problem 4

Let a_i^T denote row i of A. Write the average least squares loss as

$$J(x) = \frac{1}{2m}\|Ax - y\|^2 = \frac{1}{m}\sum_{i=1}^{m} \frac{1}{2}(a_i^T x - y_i)^2.$$

- Find the gradient of the loss associated with a single observation, $\frac{1}{2}(a_i^T x - y_i)^2$.
- For a randomly chosen minibatch B containing b observations, write down an estimate $g_B(x)$ of the gradient of $J(x)$ and a corresponding minibatch gradient update.
- What choices of b give a) stochastic gradient descent and b) standard, or full-batch, gradient descent?

- Contrast a minibatch update with the Landweber update from Problem 2.

Problem 5

A training set has 200 observations. At the start of each epoch it is randomly divided into 20 non-overlapping minibatches of 10 observations each.

- After three epochs, how many times has the full data set been used, and how many parameter updates have occurred?
- How many updates would occur per epoch if the batch size were a) one and b) 200?
- Give one computational advantage and one possible disadvantage of using minibatches rather than the full batch.
- Does completing more epochs necessarily give a better reconstruction for a noisy inverse problem? Relate your answer to semi-convergence, early stopping, and the choice of learning rate.

Lecture 13: Regularisation parameter choice

Problem 6

Suppose the data are described by the statistical model

$$Y = Ax^* + E,$$

where the entries E_i are independent, have mean zero, and have variance σ^2. You may use

$$\mathrm{Var}(Z) = \mathbb{E}\left[(Z - \mathbb{E}[Z])^2\right] = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2,$$

and the linearity of expectation.

- Show that

$$\mathbb{E}\left[\|Y - Ax^*\|^2\right] = m\sigma^2.$$

- Suppose $m = 50$ and $\sigma = 0.2$. Compute the expected squared data misfit for the true model. What might squared residuals of 0.05, 2.1, and 20 respectively suggest about three fitted models?
- Explain how the discrepancy principle uses this information to choose a regularisation parameter. Why can it be described as choosing a simple model that is still compatible with the noise level?
- What happens to the selected level of regularisation if σ is substantially under-estimated? What if it is over-estimated?

Problem 7

Let a_i^T again denote row i of A. For each i, let $x_\lambda^{(-i)}$ be the Tikhonov solution obtained after leaving observation (a_i^T, y_i) out of the fitting data.

- Write down the leave-one-out cross-validation score for λ.
- Explain why the prediction of y_i must be formed using $x_\lambda^{(-i)}$, rather than a solution fitted using y_i itself.
- Describe how cross-validation uses these scores to choose λ, and what should be done after λ has been chosen.
- Why would choosing λ using only the training data misfit tend to favour too little regularisation?
- What is generalised cross-validation intended to approximate, and why can it be computationally attractive for linear fitting rules?

Problem 8

Describe and contrast the following methods for choosing a regularisation parameter:

- the L-curve method;
- the discrepancy principle;
- cross-validation.

For each method, state what information it requires, what quantity or trade-off it examines, and how the parameter is selected. Would you expect the three methods always to select the same value? Explain briefly.

Lecture 14: Hybrid physics/machine learning models

Problem 9

The lecture described several places where physical or scientific knowledge can enter a data-driven model. Classify each of the following as placing the physics mainly in the *forward model*, the *loss or regularisation term*, the *training data*, or the *model architecture*:

- a differential-equation simulator used to map unknown parameters to predicted observations;
- a penalty on the residual of a differential equation;
- a neural-network surrogate trained on outputs generated by a physical simulator;
- a neural-network architecture constructed so that it conserves mass exactly.

Which of these approaches impose the physics as a hard constraint, which allow an explicit trade-off with the data, and which do neither directly? Briefly discuss what could happen if the physical model is incorrect.

Problem 10

Suppose a time-dependent function is represented on a fine grid by the basis expansion

$$y = B\beta,$$

where $B \in \mathbb{R}^{n \times p}$. Observations are made at only m of the grid points, so that

$$y_{\text{obs}} = A_{\text{obs}} B\beta + e,$$

where $A_{\text{obs}} \in \mathbb{R}^{m \times n}$ is an observation or selection matrix. For vertical projectile motion, suppose the physical model is $y'' = -g$, represented on the grid by

$$D_2 B\beta \approx -g\mathbf{1}.$$

- Formulate a regularised least squares problem that balances fitting the observed data with approximately satisfying the physical model.
- Rewrite your problem as a standard augmented least squares problem. Give the augmented matrix and right-hand side.
- Explain why the appropriate penalty contains $D_2 B\beta$, rather than $D_2\beta$.
- Describe the fitted solutions in the limits $\lambda \to 0$ and $\lambda \to \infty$. What trade-off would you expect if the true projectile experienced air resistance, but the regularisation term assumed gravity only?

Problem 11

Replace the basis expansion in Problem 10 by a neural network $h_\theta(t)$ representing the projectile height.

- Write down a physics-informed neural network (PINN) loss containing a data-fitting term at the observation times and a physics term enforcing $h''_\theta(t) = -g$ at a set of fine-grid or collocation points.
- How can $h''_\theta(t)$ be calculated when training the network?
- What role does the relative weight between the two terms play? Relate this to regularisation parameter choice.
- Contrast solving this problem using a spline or other linear basis expansion with solving it using a neural network.